

Functional Methods & Grassmann Integrals — Osborn AQFT Example Sheet 2

H. Osborn, Advanced Quantum Field Theory, Example Sheet 2 (functional methods, effective action, Grassmann integrals, SUSY QM).

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Quantum Field Theory

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1. Legendre Transform and Perturbative Expansion, Q1

Let, with $x_i, b_i, a_i \in \mathbb{R}^n$,

$$e^{W(b)} = \int d^n x e^{-S+b \cdot x}, \quad S = \frac{1}{2} x \cdot \mathbf{A} x + V(x), \quad \frac{\partial}{\partial b_i} W(b) = a_i, \quad \Gamma(a) + W(b) = b \cdot a. \quad (1.1)$$

Show that

$$\frac{\partial}{\partial a_i} \Gamma(a) = b_i, \quad W''_{ij} \Gamma''_{jk} = \delta_{ik}, \quad \frac{\partial}{\partial b_i} = W''_{ij} \frac{\partial}{\partial a_j}, \quad (1.2)$$

where W'' , Γ'' are matrices formed by $\frac{\partial^2}{\partial b_i \partial b_j} W$ and $\frac{\partial^2}{\partial a_i \partial a_j} \Gamma$. Find the first few terms in the perturbative expansion of $W(b)$ and correspondingly for $\Gamma(a)$.

Solution.

$$\frac{\partial}{\partial a_i} (\Gamma(a) + W(b)) = \frac{\partial}{\partial a_i} \Gamma(a) = b_i. \quad (1.3)$$

$$W''_{ij} \Gamma''_{jk} = \frac{\partial a_i}{\partial b_j} \frac{\partial b_j}{\partial a_k} = \delta_{ik}. \quad (1.4)$$

$$\frac{\partial}{\partial b_i} = \frac{\partial a_j}{\partial b_i} \frac{\partial}{\partial a_j} = W''_{ij} \frac{\partial}{\partial a_j}. \quad (1.5)$$

Now, Taylor expands the zero-dimensional Euclidean path integral, we find

$$\begin{aligned} e^{W(b)} &= e^{-V(\partial/\partial b)} \int_{\mathbb{R}^n} d^n x e^{-\frac{1}{2} (x_i - A_{il}^{-1} b_l) A_{ij} (x_j - A_{jk}^{-1} b_k) + \frac{1}{2} b_i A_{ij}^{-1} b_j} \\ &= e^{-V(\partial/\partial b)} \int_{\mathbb{R}^n} d^n \xi e^{-\frac{1}{2} \xi_i A_{ij} \xi_j + \frac{1}{2} b_i A_{ij}^{-1} b_j} \\ &= \left(\int_{\mathbb{R}^n} d^n \xi e^{-\frac{1}{2} \xi_i A_{ij} \xi_j} \right) e^{-V(\partial/\partial b)} e^{\frac{1}{2} b_i A_{ij}^{-1} b_j} \\ &= \frac{(2\pi)^{n/2}}{\sqrt{\det A}} \sum_{j=0}^{\infty} \frac{(-1)^j}{j!} \left[V \left(\frac{\partial}{\partial b} \right) \right]^j e^{\frac{1}{2} b_i A_{ij}^{-1} b_j}. \end{aligned} \quad (1.6)$$

where we have made the substitution $\xi_i = x_i - A_{ij}^{-1} b_j$. Now, we shall take natural log on both side, which gives

$$W(b) = \ln \left(\frac{(2\pi)^{n/2}}{\sqrt{\det A}} \right) + \ln \left\{ \left(1 + \sum_{j=1}^{\infty} \frac{(-1)^j}{j!} \left[V \left(\frac{\partial}{\partial b} \right) \right]^j \right) e^{\frac{1}{2} b_i A_{ij}^{-1} b_j} \right\}. \quad (1.7)$$

Now, given

$$e^{-\frac{1}{2}b_i A_{ij}^{-1} b_j} \frac{\partial}{\partial b_i} e^{\frac{1}{2}b_i A_{ij}^{-1} b_j} = \frac{\partial}{\partial b_i} + A_{ij}^{-1} b_j, \quad (1.8)$$

we have

$$e^{-\frac{1}{2}b_i A_{ij}^{-1} b_j} V \left(\frac{\partial}{\partial b} \right) e^{\frac{1}{2}b_i A_{ij}^{-1} b_j} = V \left(\frac{\partial}{\partial b} + A^{-1} b \right). \quad (1.9)$$

Therefore,

$$W(b) = W_0(0) + \frac{1}{2} b_i A_{ij}^{-1} b_j + \sum_{l=1}^{\infty} \frac{1}{l} \sum_{j=1}^{\infty} \frac{(-1)^{j+2l+1}}{(j!)^l} \left[V \left(\frac{\partial}{\partial b} + A^{-1} b \right) \right]^{j+l}, \quad (1.10)$$

where the derivative is understood to act on 1.

Let us take the potential to be

$$V(x) = \frac{\lambda}{4!} x^4, \quad (1.11)$$

the connected generating function becomes

$$W(b) = W_0(0) + \frac{1}{2} b_i A_{ij}^{-1} b_j + \sum_{n=1}^{\infty} \frac{1}{n} \left\{ \sum_{m=1}^{\infty} \frac{(-1)^{m+1} \lambda^m}{m! 4!} \left[\left(\frac{\partial}{\partial b_i} + A_{ij}^{-1} b_j \right)^4 \right]^m \right\}^n. \quad (1.12)$$

Now, let us compute the first few terms of the above,

$$\begin{aligned} \left(\frac{\partial}{\partial b_i} + A_{ij}^{-1} b_j \right)^4 &= \left(\frac{\partial}{\partial b_i} + A_{ij}^{-1} b_j \right)^3 A_{ik}^{-1} b_k \\ &= \left(\frac{\partial}{\partial b_i} + A_{ij}^{-1} b_j \right)^2 (A_{ii}^{-1} + A_{ik}^{-1} A_{il}^{-1} b_l b_k) \\ &= \left(\frac{\partial}{\partial b_i} + A_{ij}^{-1} b_j \right) (3A_{ii}^{-1} A_{ip}^{-1} b_p + A_{ik}^{-1} A_{il}^{-1} A_{ip}^{-1} b_l b_k b_p) \\ &= (3A_{ii}^{-1} A_{ii}^{-1} + 6A_{ii}^{-1} A_{ij}^{-1} A_{il}^{-1} b_j b_l + A_{ik}^{-1} A_{il}^{-1} A_{ip}^{-1} A_{iq}^{-1} b_l b_k b_p b_q), \end{aligned} \quad (1.13)$$

which corresponds to the diagrams

$$\frac{\lambda}{4!} \left(\frac{\partial}{\partial b_i} + A_{ij}^{-1} b_j \right)^4 = \frac{1}{8} \text{ (two circles joined at a point)} + \frac{1}{4} b \text{ --- } \text{ (circle on a line)} b + \frac{1}{24} \text{ (X diagram)} . \quad (1.14)$$

Therefore, to one-loop order, we have

$$W(b) = W_0 + \frac{1}{2} b \text{ --- } b - \frac{1}{8} \text{ (two circles joined at a point)} - \frac{1}{4} b \text{ --- } \text{ (circle on a line)} b - \frac{1}{24} \text{ (X diagram)} + O(\lambda^2). \quad (1.15)$$

The quantum action can be obtained by performing a Legendre transformation, which is

$$\Gamma(a) = -W(b) + b \cdot a, \quad (1.16)$$

where

$$a_i = \frac{\partial W(b)}{\partial b_i} = A_{ij}^{-1} b_j - \frac{1}{2} \lambda A_{il}^{-1} A_{li}^{-1} A_{lj}^{-1} b_j - \frac{1}{6} \lambda A_{li}^{-1} A_{lj}^{-1} A_{lk}^{-1} A_{lp}^{-1} b_j b_k b_p + \mathcal{O}(\lambda^2). \quad (1.17)$$

Therefore, to first order in λ , we can replace b_i with $A_{ij} a_j$

$$\Gamma(a) = -W_0(0) + \frac{1}{2} A_{ij} a_i a_j + \frac{\lambda}{8} (A_{ii}^{-1})^2 + \frac{\lambda}{4} A_{ii}^{-1} a_i^2 + \frac{\lambda}{24} a_i^4 + \mathcal{O}(\lambda^2). \quad (1.18)$$

2. Effective Action for Free Scalar Field, Q2

Starting from $Z[J]$, determine $W[J]$ and $\Gamma[\phi]$ for the free field theory for a scalar field $\phi(x)$ of mass m . What are

$$\left. \frac{\delta^2}{\delta J(x) \delta J(y)} W[J] \right|_{J=0}, \quad \left. \frac{\delta^2}{\delta \phi(x) \delta \phi(y)} \Gamma[\phi] \right|_{\phi=0} \quad (2.1)$$

in this case? Verify that they satisfy the expected inverse relations.

Solution. For a free scalar field theory,

$$Z[J] = \int \mathcal{D}\phi e^{iS[\phi] + i \int d^d x J \phi}, \quad (2.2)$$

where

$$S[\phi] = \int d^d x \left[\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} (m^2 - i\epsilon) \phi^2 \right]. \quad (2.3)$$

We define

$$\phi_J(x) = \phi(x) - i \int d^d y D_F(x-y) J(y), \quad (2.4)$$

where

$$D_F(x-y) = \int \frac{d^d k}{(2\pi)^d} \frac{i e^{-ik \cdot (x-y)}}{k^2 - m^2 + i\epsilon}. \quad (2.5)$$

The exponent of the generating functional becomes

$$\begin{aligned} & -\frac{i}{2} \int d^d x \phi(x) (\square + m^2 - i\epsilon) \phi(x) + i \int d^d x J(x) \phi(x) \\ &= -\frac{i}{2} \int d^d x \phi_J(x) (\square + m^2 - i\epsilon) \phi_J(x) - \frac{1}{2} \int d^d x \int d^d y J(x) D_F(x-y) J(y). \end{aligned} \quad (2.6)$$

Therefore, we have

$$Z[J] = \exp \left[-\frac{1}{2} \int d^d x \int d^d y J(x) D_F(x-y) J(y) \right] \frac{(2\pi)^{\infty/2}}{\sqrt{\text{Det}(m^2 - \bar{\square})}}, \quad (2.7)$$

where $\bar{\square} = \partial_\tau^2 + \nabla^2$ is the Euclidean Laplace operator. Then, the connected diagram generating functional is given by

$$iW[J] = \ln Z[J] - \frac{1}{2} \int d^d x \int d^d y J(x) D_F(x-y) J(y). \quad (2.8)$$

The quantum action is defined by the Legendre transformation

$$\Gamma[\phi_J] = -W[J] + \int d^d x J(x) \phi_J(x), \quad (2.9)$$

where

$$\frac{\delta \Gamma[\phi_J]}{\delta \phi_J(x)} = J(x), \quad \frac{\delta W[J]}{\delta J(x)} = \phi_J(x). \quad (2.10)$$

Therefore,

$$\phi_J(x) = i \int d^d y D_F(x-y) J(y), \quad (2.11)$$

and hence

$$\Gamma[\phi_J] = i \ln Z[0] + \frac{1}{2} \int d^d x \phi_J(x) (\square + m^2 - i\epsilon) \phi_J(x). \quad (2.12)$$

Now, we compute

$$\left. \frac{\delta^2}{\delta J(x) \delta J(y)} W[J] \right|_{J=0} = i D_F(x-y), \quad \left. \frac{\delta^2}{\delta \phi(x) \delta \phi(y)} \Gamma[\phi] \right|_{\phi=0} = (\square + m^2 - i\epsilon), \quad (2.13)$$

which verifies

$$\left. \frac{\delta^2}{\delta J(x) \delta J(y)} W[J] \right|_{J=0} \left. \frac{\delta^2}{\delta \phi(y) \delta \phi(z)} \Gamma[\phi] \right|_{\phi=0} = \delta^d(x-z). \quad (2.14)$$

3. Relations Between Green's Functions and 1PI Vertices, Q3

Starting from the definition of the effective action $\Gamma[\phi]$ in terms of the generating functional for connected graphs $W[J]$,

$$W[J] + \Gamma[\phi] = \int d^d x J \phi, \quad \frac{\delta}{\delta J(x)} W[J] = \phi(x), \quad (3.1)$$

obtain

$$-\int d^d z G_2(x, z) \Gamma_2(z, y) = \delta^d(x - y), \quad -i \frac{\delta}{\delta J(x)} = \int d^d y G_2(x, y) \frac{\delta}{\delta \phi(y)}, \quad (3.2)$$

where

$$G_n(x_1, \dots, x_n) = (-i)^{n-1} \frac{\delta}{\delta J(x_1)} \cdots \frac{\delta}{\delta J(x_n)} W[J], \quad \Gamma_n(x_1, \dots, x_n) = -i \frac{\delta}{\delta \phi(x_1)} \cdots \frac{\delta}{\delta \phi(x_n)} \Gamma[\phi]. \quad (3.3)$$

Show that

$$G_3(x_1, x_2, x_3) = \int d^d y_1 d^d y_2 d^d y_3 G_2(x_1, y_1) G_2(x_2, y_2) G_2(x_3, y_3) \Gamma_3(y_1, y_2, y_3). \quad (3.4)$$

Also obtain relations in diagrammatic form for G_4 , G_5 in terms of G_2 , Γ_3 , Γ_4 and Γ_5 .

Solution.

$$\frac{\delta \phi(x)}{\delta \phi(y)} = \delta^d(x - y) = \int d^d z \frac{\delta \phi(x)}{\delta J(z)} \frac{\delta J(z)}{\delta \phi(y)}. \quad (3.5)$$

Recall

$$\frac{\delta \Gamma[\phi]}{\delta \phi(x)} = J(x), \quad \frac{\delta W[J]}{\delta J(x)} = \phi(x), \quad (3.6)$$

which gives

$$\delta^d(x - y) = \int d^d z \frac{\delta^2 W[J]}{\delta J(x) \delta J(z)} \frac{\delta^2 \Gamma[\phi]}{\delta \phi(z) \delta \phi(y)} = - \int d^d z G_2(x, z) \Gamma_2(z, y). \quad (3.7)$$

Similarly,

$$-i \frac{\delta}{\delta J(x)} = -i \int d^d y \frac{\delta \phi(y)}{\delta J(x)} \frac{\delta}{\delta \phi(y)} = \int d^d y G_2(x, y) \frac{\delta}{\delta \phi(y)}. \quad (3.8)$$

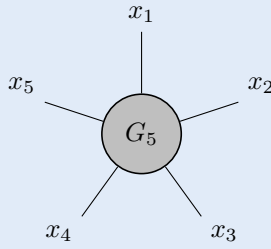
Now, we turn to the exact connected three point function

$$\begin{aligned} G_3(x_1, x_2, x_3) &= - \int d^d y_1 d^d y_2 d^d y_3 \left(i G_2(x_1, y_1) \frac{\delta}{\delta \phi(y_1)} \right) \left(i G_2(x_2, y_2) \frac{\delta}{\delta \phi(y_2)} \right) \\ &\quad \left(i G_2(x_3, y_3) \frac{\delta}{\delta \phi(y_3)} \right) \left(-\Gamma[\phi] + \int d^d x J(x) \phi(x) \right) \\ &= \int d^d y_1 d^d y_2 d^d y_3 G_2(x_1, y_1) G_2(x_2, y_2) G_2(x_3, y_3) \Gamma_3(y_1, y_2, y_3), \end{aligned} \quad (3.9)$$

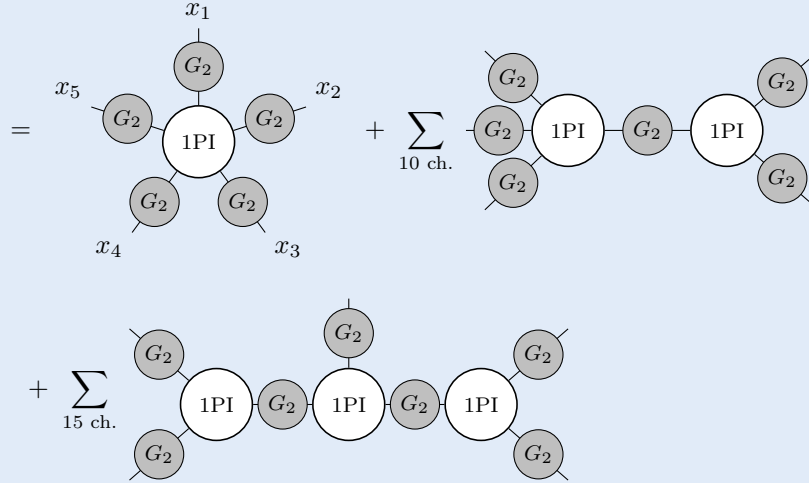
where we have jumped a few lines of algebra.

For higher-order connected correlation functions, we have the diagrammatic expressions:

$$G_4(x_1, x_2, x_3, x_4) = \text{1PI}(x_1, x_2, x_3, x_4) + \sum_{(ij)(kl)} G_2(x_i, x_j) \text{1PI}(x_i, x_j, x_k, x_l) G_2(x_k, x_l) \quad (3.10)$$



$$(3.11)$$



$$(3.12)$$

4. Grassmann Gaussian Integral with Bosonic Background, Q4

Evaluate, for $\theta_i, \bar{\theta}_i$ Grassmann variables,

$$Z = \frac{1}{(2\pi)^{n/2}} \int d^n x \prod_{i=1}^n d\bar{\theta}_i d\theta_i e^{-S(x, \theta, \bar{\theta})}, \quad S(x, \theta, \bar{\theta}) = \frac{1}{2} w_i(x) w_i(x) + \bar{\theta}_i w_{i,j}(x) \theta_j, \quad (4.1)$$

where $w_{i,j}(x) = \frac{\partial}{\partial x_j} w_i(x)$ and assuming $\det[w_{i,j}(x)] > 0$.

If $w_i(x) = x_i + \frac{1}{2} g_{ijk} x_j x_k$, $g_{ijk} = g_{ikj}$, what are the Feynman rules for a perturbative expansion? Work out the two-loop contributions to $Z(g)$ (this is much simpler if g_{ijk} is completely symmetric).

If $w_i(x) \rightarrow \lambda w_i(x)$ in S , show that

$$\frac{d}{d\lambda} e^{-S(x, \theta, \bar{\theta})} = - \left(\theta_j \frac{\partial}{\partial x_j} + \lambda w_j(x) \frac{\partial}{\partial \theta_j} \right) \bar{\theta}_i w_{i,j}(x) e^{-S(x, \theta, \bar{\theta})}. \quad (4.2)$$

By letting $x_i \rightarrow \lambda x_i$, $g_{ijk} \rightarrow g_{ijk}/\lambda$, what does this imply for $Z(g)$?

Solution. Let us first evaluate the Grassmann integral by making the substitution $\theta'_i = w_{i,j} \theta_j$

$$\int \prod_{k=1}^n d\bar{\theta}_k d\theta_k e^{-\bar{\theta}_i w_{i,j} \theta_j} = \prod_{k=1}^n \frac{\partial}{\partial \bar{\theta}_k} \frac{\partial}{\partial \theta_k} (1 - \bar{\theta}_i w_{i,j} \theta_j) = \det w_{i,j} \left(\frac{\partial}{\partial \bar{\theta}} \frac{\partial}{\partial \theta'} (1 - \bar{\theta} \theta') \right)^n = \det w_{i,j}. \quad (4.3)$$

Then, we make the substitution $d^n x = d^n w / \det w_{i,j}$, which gives

$$Z = \frac{1}{(2\pi)^{n/2}} \int d^n w e^{-w_i w_i/2} = 1. \quad (4.4)$$

For $w_i(x) = x_i + \frac{1}{2} g_{ijk} x_j x_k$, $w_{i,j}(x) = \delta_{ij} + g_{ijk} x_k$ the action becomes

$$S = \frac{1}{2} x_i x_i + \frac{1}{2} g_{ijk} x_i x_j x_k + \frac{1}{8} g_{ijk} g_{ilm} x_j x_k x_l x_m + \bar{\theta}_i \theta_i + g_{ijk} \bar{\theta}_i x_j \theta_k. \quad (4.5)$$

We can read off the Feynman rules:

$$i \text{-----} j = \delta_{ij}, \quad (4.6)$$

$$i \longrightarrow j = \delta_{ij}, \quad (4.7)$$

$$\begin{array}{c} j \\ \diagup \\ i \text{-----} \bullet \\ \diagdown \\ k \end{array} = -g_{ijk}, \quad (4.8)$$

$$\begin{array}{c} j \quad k \\ \diagdown \quad \diagup \\ \times \\ \diagup \quad \diagdown \\ l \quad m \end{array} = -g_{ijk}g_{ilm}, \quad (4.9)$$

$$\begin{array}{c} j \\ | \\ i \longrightarrow \bullet \longrightarrow k \end{array} = -g_{ijk}. \quad (4.10)$$

Since we have already shown that $Z = 1$, the loop expansions should be cancelled order by order in g . There are no diagrams of order g . For order g^2 , the diagram should cancel, but we shall not show it explicitly here.

Now, we take $w_i \rightarrow \lambda w_i$, which the action becomes

$$S = \frac{1}{2} \lambda^2 w_i(x) w_i(x) + \lambda \bar{\theta}_i w_{i,j}(x) \theta_j. \quad (4.11)$$

Then,

$$\frac{d}{d\lambda} e^{-S(x,\theta,\bar{\theta})} = -(\lambda w_i(x) w_i(x) + \bar{\theta}_i w_{i,j}(x) \theta_j) e^{-S(x,\theta,\bar{\theta})}, \quad (4.12)$$

while

$$\begin{aligned} & - \left(\theta_j \frac{\partial}{\partial x_j} + \lambda w_j(x) \frac{\partial}{\partial \theta_j} \right) \bar{\theta}_i w_i(x) e^{-S(x,\theta,\bar{\theta})} \\ &= - \left(\theta_j w_{i,j}(x) \bar{\theta}_i - \theta_j \bar{\theta}_i w_i(x) [\lambda^2 w_k(x) w_{k,j}(x) + \lambda \bar{\theta}_k w_{i,lj}(x) \theta_l] \right. \\ & \quad \left. + \lambda w_i(x) w_i(x) - \lambda^2 \bar{\theta}_i w_j(x) w_i(x) w_{l,j}(x) \theta_l \right) e^{-S(x,\theta,\bar{\theta})} \\ &= - (\lambda w_i(x) w_i(x) + \bar{\theta}_i w_{i,j}(x) \theta_j) e^{-S(x,\theta,\bar{\theta})}, \end{aligned} \quad (4.13)$$

where the term containing $w_{i,jl}$ vanishes by anti-commutation of Grassmann numbers. Letting $x_i \rightarrow \lambda x_i$ and $g_{ijk} \rightarrow g_{ijk}/\lambda$ is equivalent to letting $w_i \rightarrow \lambda w_i$. In other words,

$$w_i(\lambda x, g/\lambda) = \lambda w_i(x). \quad (4.14)$$

5. Fermionic Gaussian Integral, Q5

Show that, for $\theta_i, \bar{\theta}_i, \eta_i, \bar{\eta}_i, i = 1, 2, \dots, n$ independent Grassmann variables and B an invertible $n \times n$ matrix,

$$\int \prod_{i=1}^n d\bar{\theta}_i d\theta_i \exp(-\bar{\theta}_i B_{ij} \theta_j + \bar{\eta}_i \theta_i + \bar{\theta}_i \eta_i) = \det B \exp(\bar{\eta}_i (B^{-1})_{ij} \eta_j), \quad (5.1)$$

and

$$\langle \theta_i \bar{\theta}_j \rangle = -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} \exp(\bar{\eta}_i (B^{-1})_{ij} \eta_j) \Big|_{\eta=\bar{\eta}=0} = (B^{-1})_{ij}. \quad (5.2)$$

Solution. Making the substitution $\chi_i = \theta_i - B_{ij}^{-1} \eta_j$, and $\bar{\chi}_i = \bar{\theta}_i - \bar{\eta}_j B_{ji}^{-1}$, the integral becomes

$$\int \prod_{i=1}^n d\bar{\chi}_i d\chi_i \exp(-\bar{\chi}_i B_{ij} \chi_j + \bar{\eta}_j B_{ji}^{-1} \eta_i) = \exp(\bar{\eta}_i (B^{-1})_{ij} \eta_j) \int \prod_{i=1}^n d\bar{\chi}_i d\chi_i \exp(-\bar{\chi}_i B_{ij} \chi_j), \quad (5.3)$$

where the latter integral is equal to $\det B$. Then,

$$\langle \theta_i \bar{\theta}_j \rangle = -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} \exp(\bar{\eta}_i (B^{-1})_{ij} \eta_j) \Big|_{\eta=\bar{\eta}=0} = -\frac{\partial}{\partial \bar{\eta}_i} \frac{\partial}{\partial \eta_j} (1 + \bar{\eta}_i B_{ij}^{-1} \eta_j) \Big|_{\eta=\bar{\eta}=0} = B_{ij}^{-1}. \quad (5.4)$$

6. Pfaffian and Real Grassmann Integrals, Q6

Show that

$$\int d^{2n} \theta \exp\left(\frac{1}{2} \theta_i A_{ij} \theta_j + \eta_i \theta_i\right) = \text{Pf}(A) \exp\left(\frac{1}{2} \eta_i (A^{-1})_{ij} \eta_j\right), \quad d^{2n} \theta = d\theta_{2n} \cdots d\theta_1, \quad (6.1)$$

where θ_i and η_i are Grassmann variables, A is an invertible antisymmetric matrix and the Pfaffian is

$$\text{Pf}(A) = \frac{1}{2^n n!} \varepsilon_{i_1 \dots i_{2n}} A_{i_1 i_2} \cdots A_{i_{2n-1} i_{2n}}. \quad (6.2)$$

Why does $\text{Pf}(A)^2 = \det A$? Hence obtain

$$\langle \theta_i \theta_j \rangle = \frac{\partial}{\partial \eta_i} \frac{\partial}{\partial \eta_j} \exp\left(\frac{1}{2} \eta_i (A^{-1})_{ij} \eta_j\right) \Big|_{\eta=0} = -(A^{-1})_{ij}. \quad (6.3)$$

Show that $\delta(\theta) = \theta_1 \cdots \theta_{2n}$ plays the role of a δ -function for Grassmann integrals over $\theta_1, \dots, \theta_{2n}$.

Solution. Similarly, we perform the substitution $\chi_i = \theta_i - A_{ij}^{-1} \eta_j$, which gives

$$\int d^{2n} \chi \exp\left(\frac{1}{2} (\chi_i + A_{il}^{-1} \eta_l) A_{ij} (\chi_j + A_{jk}^{-1} \eta_k) + \eta_i (\chi_i + A_{ij}^{-1} \eta_j)\right). \quad (6.4)$$

The exponent is

$$\frac{1}{2} \chi_i A_{ij} \chi_j + \frac{1}{2} \chi_i A_{ij} A_{jk}^{-1} \eta_k + \frac{1}{2} A_{il}^{-1} \eta_l A_{ij} \chi_j + \frac{1}{2} A_{il}^{-1} \eta_l A_{ij} A_{jk}^{-1} \eta_k + \eta_i \chi_i + \eta_i A_{ij}^{-1} \eta_j. \quad (6.5)$$

For the $\theta A \theta$ term to be non-vanishing, we need A to be antisymmetric. Therefore,

$$\frac{1}{2} A_{il}^{-1} \eta_l A_{ij} \chi_j = -\frac{1}{2} \eta_i \chi_i. \quad (6.6)$$

The integral becomes

$$\int d^{2n} \theta \exp\left(\frac{1}{2} \theta_i A_{ij} \theta_j + \eta_i \theta_i\right) = \exp\left(\frac{1}{2} \eta_i (A^{-1})_{ij} \eta_j\right) \int d^{2n} \chi \exp\left(\frac{1}{2} \chi_i A_{ij} \chi_j\right). \quad (6.7)$$

The integral on the right-hand side is

$$\begin{aligned} \int d^{2n} \chi \exp \left(\frac{1}{2} \chi_i A_{ij} \chi_j \right) &= \frac{\partial}{\partial \chi_{2n}} \cdots \frac{\partial}{\partial \chi_1} \sum_{m=0}^n \frac{1}{m!} \left(\frac{1}{2} \chi_i A_{ij} \chi_j \right)^m \\ &= \frac{1}{2^n n!} \frac{\partial}{\partial \chi_{2n}} \cdots \frac{\partial}{\partial \chi_1} (\chi_i A_{ij} \chi_j)^n \\ &= \frac{1}{2^n n!} \varepsilon_{i_1 \cdots i_{2n}} A_{i_1 i_2} \cdots A_{i_{2n-1} i_{2n}} \\ &= \text{Pf}(A). \end{aligned} \quad (6.8)$$

Now, we consider

$$\begin{aligned} \text{Pf}^2(A) &= \left(\int d^{2n} \chi \exp \left(\frac{1}{2} \chi_i A_{ij} \chi_j \right) \right)^2 \\ &= \int d^{2n} \chi d^{2n} \chi' \exp \left(\frac{1}{2} \chi_i A_{ij} \chi_j + \frac{1}{2} \chi'_i A_{ij} \chi'_j \right). \end{aligned} \quad (6.9)$$

Let $\eta_i = \frac{1}{\sqrt{2}}(\chi_i + \chi'_i)$ and $\eta'_i = \frac{i}{\sqrt{2}}(\chi_i - \chi'_i)$, so that the above becomes

$$\text{Pf}^2(A) = \int \prod_{i=1}^{2n} d\eta_i d\eta'_i \exp(\eta'_i A_{ij} \eta_j) = \det(-A) = (-1)^{2n} \det(A) = \det(A). \quad (6.10)$$

It is easy to see that

$$\langle \theta_i \theta_j \rangle = \frac{\partial}{\partial \eta_i} \frac{\partial}{\partial \eta_j} \exp \left(\frac{1}{2} \eta_i (A^{-1})_{ij} \eta_j \right) \Big|_{\eta=0} = -(A^{-1})_{ij} \quad (6.11)$$

is true.

Now, consider a Grassmann function, which can be expanded as $f(\theta) = f(0) + (\text{terms with } \theta_i)$. Therefore,

$$\int d^{2n} \theta \theta_1 \cdots \theta_{2n} f(\theta) = \int d^{2n} \theta \theta_1 \cdots \theta_{2n} f(0) = f(0). \quad (6.12)$$

7. Fermionic Path Integral and Partition Function, Q7

For the simple fermionic oscillator with energy eigenstates $|0\rangle$, $|1\rangle$ and $H = b^\dagger b \omega$, let $|\theta\rangle = |0\rangle + \theta|1\rangle$, $\langle \bar{\theta}| = \langle 0| + \bar{\theta}\langle 1|$ where $\theta, \bar{\theta}$ are Grassmann variables. Show that

$$\text{tr} e^{-\beta H} = \int d\theta d\bar{\theta} e^{\theta \bar{\theta}} \langle \bar{\theta} | e^{-\beta H} | \theta \rangle, \quad \int d\bar{\theta} d\theta e^{\theta \bar{\theta}} |\theta\rangle \langle \bar{\theta}| = 1. \quad (7.1)$$

Let $\beta = (N+1)\epsilon$ and show that this can be expressed for large N as a path integral over products of $\langle \bar{\theta}_{i+1} | e^{-H} | \theta_i \rangle \approx \exp((1 - \epsilon\omega)\bar{\theta}_{i+1}\theta_i)$ so that

$$\text{tr} e^{-\beta H} \approx \int \prod_{i=1}^{N+1} d\bar{\theta}_i d\theta_i \exp(-\bar{\theta}_i B_{ij} \theta_j) = \det(B), \quad (7.2)$$

where we take $\bar{\theta}_{N+1} = \bar{\theta}$, $\theta_{N+1} = -\theta$ and B is an $(N+1) \times (N+1)$ matrix of the form

$$B = \begin{pmatrix} 1 & 0 & \cdots & 0 & 1 - \epsilon\omega \\ -1 + \epsilon\omega & 1 & \cdots & 0 & 0 \\ 0 & -1 + \epsilon\omega & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & -1 + \epsilon\omega & 1 \end{pmatrix}. \quad (7.3)$$

Show that $\det B = 1 + (1 - \epsilon\omega)^{N+1} \rightarrow 1 + e^{-\beta\omega}$ as $N \rightarrow \infty$.

Solution. The completeness relation is given by

$$|0\rangle\langle 0| + |1\rangle\langle 1| = 1 = \int d\bar{\theta} d\theta e^{\theta\bar{\theta}} |\theta\rangle\langle\bar{\theta}|. \quad (7.4)$$

Then

$$\begin{aligned} \text{tr } e^{-\beta H} &= \langle 0| e^{-\beta H} |0\rangle + \langle 1| e^{-\beta H} |1\rangle \\ &= \int d\bar{\theta} d\theta e^{\theta\bar{\theta}} (\langle\bar{\theta}| e^{-\beta H} |0\rangle + \theta \langle\bar{\theta}| e^{-\beta H} |1\rangle) \\ &= \int d\bar{\theta} d\theta e^{\theta\bar{\theta}} \langle\bar{\theta}| e^{-\beta H} |\theta\rangle. \end{aligned} \quad (7.5)$$

Taking $\beta = N + 1$, so that the path integral becomes

$$\text{tr } e^{-\beta H} = \int d\bar{\theta} d\theta e^{\theta\bar{\theta}} \int \prod_{i=1}^N (d\bar{\theta}_i d\theta_i e^{\theta_i \bar{\theta}_i}) \langle\bar{\theta}| e^{-H} |\theta_1\rangle \langle\bar{\theta}_1| e^{-H} |\theta_2\rangle \cdots \langle\bar{\theta}_N| e^{-H} |\theta\rangle. \quad (7.6)$$

For large N ,

$$\langle\bar{\theta}_{i+1}| e^{-\epsilon H} |\theta_i\rangle = \langle\bar{\theta}_{i+1}| e^{-\epsilon\omega b^\dagger b} |\theta_i\rangle \approx e^{-\epsilon\omega \bar{\theta}_{i+1}\theta_i} \langle\bar{\theta}_{i+1}| \theta_i\rangle = e^{-\epsilon\omega \bar{\theta}_{i+1}\theta_i} (1 + \bar{\theta}_{i+1}\theta_i) = e^{(1-\epsilon\omega)\bar{\theta}_{i+1}\theta_i}, \quad (7.7)$$

where we have neglected terms of order ϵ^2 or higher. Plugging this back into the path integral, we find

$$\text{tr } e^{-\beta H} \approx \int_{\theta_1=\theta_{N+2}} d\bar{\theta}_i d\theta_i e^{\theta_i \bar{\theta}_i} e^{(1-\epsilon\omega)\bar{\theta}_{i+1}\theta_i} = \int \prod_{i=1}^{N+1} d\bar{\theta}_i d\theta_i \exp(-\bar{\theta}_i B_{ij} \theta_j) = \det(B). \quad (7.8)$$

The determinant is given by

$$\begin{aligned} \det(B) &= \det \begin{pmatrix} 1 & \cdots & 0 & 0 \\ -1 + \epsilon\omega & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 \\ 0 & \cdots & -1 + \epsilon\omega & 1 \end{pmatrix} + (-1)^N (1 - \epsilon\omega) \det \begin{pmatrix} -1 + \epsilon\omega & 1 & \cdots & 0 \\ 0 & -1 + \epsilon\omega & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ 0 & 0 & \cdots & -1 + \epsilon\omega \end{pmatrix} \\ &= 1 + (-1)^N (1 - \epsilon\omega)(-1 + \epsilon\omega)^N = 1 + (1 - \epsilon\omega)^{N+1} \rightarrow 1 + e^{-\beta\omega}. \end{aligned} \quad (7.9)$$

8. Supersymmetric Quantum Mechanics, Q8

Let

$$S = \int_0^\beta d\tau \left(\frac{1}{2} \dot{x}^2 + \bar{\theta} \dot{\theta} + \frac{1}{2} \lambda^2 w'(x)^2 + \lambda w''(x) \bar{\theta} \theta \right). \quad (8.1)$$

Assume $x(\tau)$ and the Grassmannian $\theta(\tau)$, $\bar{\theta}(\tau)$ are all periodic with period β . Show that

$$Q(\bar{\theta} \cdot w' e^{-S}) = -\frac{\partial}{\partial \lambda} e^{-S}, \quad Q = \int_0^\beta d\tau \left(\theta \frac{\delta}{\delta x} + (\lambda w'(x) + \dot{x}) \frac{\delta}{\delta \bar{\theta}} \right), \quad \bar{\theta} \cdot w' = \int_0^\beta d\tau \bar{\theta} w'(x). \quad (8.2)$$

Hence show that

$$Z = \int \mathcal{D}x \mathcal{D}\bar{\theta} \mathcal{D}\theta e^{-S} \quad (8.3)$$

is independent of λ if $\lambda \neq 0$. Calculate Z for $\lambda w'(x) = \omega x$ if $Z = 1$ for $\omega = 0$.

Solution.

$$\begin{aligned}
Q(\bar{\theta} \cdot w' e^{-S}) &= \int_0^\beta d\tau \left(\frac{\delta}{\delta x} + (\lambda w'(x) + \dot{x}) \frac{\delta}{\delta \bar{\theta}} \right) \left(\int_0^\beta d\tau' \bar{\theta} w'(x) e^{-S} \right) \\
&= \int_0^\beta d\tau d\tau' \left(\theta(\tau) \bar{\theta}(\tau') w''(x(\tau')) \delta(\tau - \tau') + \theta(\tau) \bar{\theta}(\tau') w'(x(\tau')) \right. \\
&\quad \times \int_0^\beta d\tau'' [\ddot{x}(\tau) \delta(\tau - \tau'') - \lambda^2 w'(x(\tau'')) w''(x(\tau'')) \delta(\tau - \tau'') \\
&\quad \left. - \lambda w'''(x(\tau'')) \bar{\theta}(\tau'') \theta(\tau'') \delta(\tau - \tau'')] \right) e^{-S} \\
&\quad + \int_0^\beta d\tau d\tau' [\lambda w'(x(\tau)) + \dot{x}(\tau)] \left(\delta(\tau - \tau') w'(x(\tau')) \right. \\
&\quad \left. + \bar{\theta}(\tau') w'(x(\tau')) \int_0^\beta d\tau'' [\dot{\theta}(\tau'') \delta(\tau - \tau'') + \lambda w''(x(\tau'')) \theta(\tau'') \delta(\tau - \tau'')] \right) e^{-S} \\
&= \int_0^\beta d\tau \theta \bar{\theta} w''(x) e^{-S} \\
&\quad + \int_0^\beta d\tau d\tau' \theta(\tau) \bar{\theta}(\tau') w'(x(\tau')) [\ddot{x}(\tau) - \lambda^2 w'(x(\tau)) w''(x(\tau))] e^{-S} \\
&\quad + \int_0^\beta d\tau [\lambda w'(x) + \dot{x}] w'(x) e^{-S} \\
&\quad + \int_0^\beta d\tau d\tau' [\lambda w'(x(\tau)) + \dot{x}(\tau)] \bar{\theta}(\tau') w'(x(\tau')) [\dot{\theta}(\tau) + \lambda w''(x(\tau)) \theta(\tau)] e^{-S} \\
&= \int_0^\beta d\tau [w''(x) \bar{\theta} \theta + \lambda w'(x) w'(x) + \dot{x} w'(x)] e^{-S} \\
&= -\frac{d}{d\lambda} e^{-S},
\end{aligned} \tag{8.4}$$

where we have used the fact that $\dot{x} w'(x) = \frac{d}{d\tau} w(x)$ is a total derivative, and thus vanishes.

Now consider

$$\frac{dZ}{d\lambda} = - \int \mathcal{D}x \mathcal{D}\bar{\theta} \mathcal{D}\theta Q(\bar{\theta} \cdot w' e^{-S}), \tag{8.5}$$

which only contains boundary terms of x and $\bar{\theta}$. Therefore, we have $dZ/d\lambda = 0$, which means Z is independent of λ for SUSY QM. Now, take $\lambda w'(x) = \omega x$, and then

$$S = \int_0^\beta d\tau \left(\frac{1}{2} \dot{x}^2 + \bar{\theta} \dot{\theta} + \frac{1}{2} \omega^2 x^2 + \omega \bar{\theta} \theta \right). \tag{8.6}$$

The partition function is then given by

$$\begin{aligned}
Z &= \int \mathcal{D}x e^{-\int d\tau \frac{1}{2} (\dot{x}^2 + \omega^2 x^2)} \int \mathcal{D}\bar{\theta} \mathcal{D}\theta e^{-\int d\tau (\bar{\theta} \dot{\theta} + \omega \bar{\theta} \theta)} \\
&= \int \mathcal{D}x e^{-\int d\tau \frac{1}{2} x (-\frac{d^2}{d\tau^2} + \omega^2) x} \int \mathcal{D}\bar{\theta} \mathcal{D}\theta e^{-\int d\tau \bar{\theta} (\frac{d}{d\tau} + \omega) \theta} \\
&= \frac{\text{Det}_{\text{AP}}(\frac{d}{d\tau} + \omega)}{\sqrt{\text{Det}_{\text{P}}(-\frac{d^2}{d\tau^2} + \omega^2)}}.
\end{aligned} \tag{8.7}$$

The determinants can be evaluated as

$$\text{Det}_{\text{AP}} \left(\frac{d}{d\tau} + \omega \right) = \prod_n \lambda_n = \prod_n \left[\frac{(2n-1)\pi i}{\beta} + \omega \right] = \text{Det}_{\text{AP}} \left(\frac{d}{d\tau} \right) \prod_n \left[1 - \frac{i\omega\beta}{(2n-1)\pi} \right], \tag{8.8}$$

and

$$\text{Det}_{\text{P}} \left(-\frac{d^2}{d\tau^2} + \omega^2 \right) = \prod_n \left[\left(\frac{2\pi n}{\beta} \right)^2 + \omega^2 \right] = \text{Det}_{\text{P}} \left(-\frac{d^2}{d\tau^2} \right) \prod_n \left[1 + \left(\frac{\omega\beta}{2\pi n} \right)^2 \right]. \tag{8.9}$$

Using the normalisation, $Z = 1$ for $\omega = 0$, we find

$$Z = \prod_n \frac{1 - \frac{i\omega\beta}{(2n-1)\pi}}{\sqrt{1 + \left(\frac{\omega\beta}{2\pi n}\right)^2}}. \quad (8.10)$$